

## (a) One-hot: “mat cat mat”

|     | mat | cat | rat |
|-----|-----|-----|-----|
| mat | 1   | 0   | 0   |
| cat | 0   | 1   | 0   |
| mat | 1   | 0   | 0   |

one row per word,  
a single 1 in each

## (b) Bag of words

|               | mat | cat | rat |
|---------------|-----|-----|-----|
| “mat cat mat” | 2   | 1   | 0   |
| “cat rat rat” | 0   | 1   | 2   |

one row of counts per sentence;  
the order of the words is lost

Neither says that two words are alike: the numbers carry no meaning.